

ClassNest

UNIT-V Advanced Python Integration and Data Handling Assessment

UNIT V: C++ Integration, SQLite Data Manipulation and Data Visualization

Coverage: Importing C++ Programs in Python, Data Manipulation through SQL, Data Visualization using PyPlot

Pattern: 30 MCQs, 20 Fill-ups, 10 Simple Coding Questions. Total: 60 questions, 100 marks.

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MCQ QUESTIONS

QUESTION 1 - MCQ - 1 MARK

In the Python-C++ comparison, Python is typically described as a/an:

- A. Compiled language
- B. Interpreted language
- C. Assembly language
- D. Markup language

Answer: B

Explanation: Python is typically interpreted, while C++ is typically compiled.

QUESTION 2 - MCQ - 1 MARK

Which language is usually statically typed according to the C++ comparison?

- A. Python
- B. C++
- C. HTML
- D. CSV

Answer: B

Explanation: C++ is statically typed; Python is dynamically typed.

QUESTION 3 - MCQ - 1 MARK

Python can act as a scripting language mainly because it can:

- A. Only draw charts
- B. Integrate and communicate with other languages
- C. Only create databases
- D. Only compile C++

Answer: B

Explanation: Scripting languages are often used to integrate with other programming languages.

QUESTION 4 - MCQ - 1 MARK

Which of the following is an application of scripting languages?

- A. Automating tasks
- B. Deleting operating systems
- C. Replacing hardware
- D. Changing monitor size

Answer: A

Explanation: Scripting languages can automate tasks and glue systems together.

QUESTION 5 - MCQ - 1 MARK

Python uses automatic garbage collection, whereas C++:

- A. Does not use automatic garbage collection in the same way
- B. Cannot use variables
- C. Cannot use functions
- D. Is not a programming language

Answer: A

Explanation: The notes contrast Python garbage collection with C++ manual-style memory management.

QUESTION 6 - MCQ - 1 MARK

Importing a C++ program in a Python program is called:

- A. Plotting
- B. Wrapping
- C. Sorting
- D. Indexing

Answer: B

Explanation: Wrapping creates a Python interface to C/C++ code.

QUESTION 7 - MCQ - 1 MARK

Which interface is commonly used for interfacing Python with C programs?

- A. Python-C-API
- B. CSV module
- C. PyPlot only
- D. fetchall()

Answer: A

Explanation: Python-C-API is listed as an interface for C programs.

QUESTION 8 - MCQ - 1 MARK

Which tool is expanded as Simplified Wrapper Interface Generator?

- A. SQLite
- B. SWIG
- C. MinGW
- D. PyPlot

Answer: B

Explanation: SWIG stands for Simplified Wrapper Interface Generator.

QUESTION 9 - MCQ - 1 MARK

MinGW is mainly used for:

- A. Compiling and linking C/C++ programs on Windows
- B. Creating pie charts only
- C. Opening CSV files only
- D. Writing HTML pages

Answer: A

Explanation: MinGW provides runtime headers/tools and g++ for C/C++ compilation on Windows.

QUESTION 10 - MCQ - 1 MARK

In the command `python pycpp.py -i pali``, the ``-i`` part represents:

- A. input mode
- B. integer mode
- C. image mode
- D. insert mode

Answer: A

Explanation: The notes describe -i as input mode in the execution syntax.

QUESTION 11 - MCQ - 1 MARK

The Python library used in the notes to work with SQLite is:

- A. sqlite3
- B. matplotlib
- C. csvonly
- D. numpydb

Answer: A

Explanation: Python has a native sqlite3 module for SQLite operations.

QUESTION 12 - MCQ - 1 MARK

SQLite stores its data in:

- A. Regular database files
- B. Only printed paper
- C. Only cloud servers
- D. Image files only

Answer: A

Explanation: SQLite is an embedded relational database stored in regular files.

QUESTION 13 - MCQ - 1 MARK

Which method is used to create a connection to a SQLite database?

- A. connect()
- B. cursor() only
- C. fetchall()
- D. show()

Answer: A

Explanation: sqlite3.connect() opens or creates the database file.

QUESTION 14 - MCQ - 1 MARK

The cursor object is created by calling:

- A. connection.cursor()
- B. database.paint()
- C. plot.cursor()
- D. csv.cursor()

Answer: A

Explanation: The cursor object is created from the connection.

QUESTION 15 - MCQ - 1 MARK

In SQLite/Python, SQL commands are executed using:

- A. cursor.execute()
- B. list.append()
- C. plt.plot()
- D. field.split()

Answer: A

Explanation: The execute method runs SQL commands through the cursor.

QUESTION 16 - MCQ - 1 MARK

After INSERT/UPDATE/DELETE operations, changes should be saved using:

- A. commit()
- B. fetchone()
- C. show()
- D. ylabel()

Answer: A

Explanation: connection.commit() saves database changes.

QUESTION 17 - MCQ - 1 MARK

A cursor is mainly used to:

- A. Traverse and fetch database records
- B. Make a pie chart
- C. Compile C++ only
- D. Store CSV formulas

Answer: A

Explanation: The notes define cursor as a control structure for fetching records.

QUESTION 18 - MCQ - 1 MARK

Which SQL command is used to retrieve records from a table?

- A. SELECT
- B. CREATE only
- C. DROP only
- D. COMMIT only

Answer: A

Explanation: SELECT retrieves/fetches data from a table.

QUESTION 19 - MCQ - 1 MARK

fetchall() is used to:

- A. Fetch all rows from the result set
- B. Fetch no rows
- C. Delete a table
- D. Create a plot title

Answer: A

Explanation: fetchall() returns all rows from the database query result.

QUESTION 20 - MCQ - 1 MARK

fetchone() returns:

- A. The next row or None if no row is left
- B. All tables in database
- C. A bar chart
- D. A C++ compiler

Answer: A

Explanation: fetchone() returns the next row or None.

QUESTION 21 - MCQ - 1 MARK

A data visualization is best described as:

- A. Graphical representation of data/information
- B. Raw data without meaning
- C. A database file only
- D. A C++ header file

Answer: A

Explanation: Data visualization communicates information visually.

QUESTION 22 - MCQ - 1 MARK

Which of the following is listed as a type of data visualization?

- A. Charts
- B. Compilers
- C. Pointers only
- D. Primary keys only

Answer: A

Explanation: Charts, tables, graphs, maps, infographics and dashboards are visualization types.

QUESTION 23 - MCQ - 1 MARK

A dashboard is mainly:

- A. A unified visual display of resources/information
- B. A Python compiler
- C. A database cursor
- D. A C++ input mode

Answer: A

Explanation: Dashboard presents complex ideas in a simple visual format.

QUESTION 24 - MCQ - 1 MARK

Matplotlib is a popular Python library for:

- A. Data visualization
- B. C++ compilation only
- C. Database locking only
- D. CSV file extension

Answer: A

Explanation: Matplotlib is used to create charts and plots.

QUESTION 25 - MCQ - 1 MARK

Which plot displays data as a collection of points?

- A. Scatter plot
- B. Pie chart
- C. Transaction log
- D. SQL table

Answer: A

Explanation: Scatter plot shows data as points using two-dimensional values.

QUESTION 26 - MCQ - 1 MARK

A box plot summarizes distribution based on:

- A. Five-number summary
- B. Only one number
- C. Only file name
- D. Only column color

Answer: A

Explanation: Box plot uses minimum, first quartile, median, third quartile and maximum.

QUESTION 27 - MCQ - 1 MARK

Which chart is suitable to show parts of a whole?

- A. Pie chart
- B. Compiler chart
- C. Cursor chart
- D. Header chart

Answer: A

Explanation: Pie charts represent proportional parts of a whole.

QUESTION 28 - MCQ - 1 MARK

Which chart commonly compares values across categories?

- A. Bar chart
- B. GCC chart
- C. SQL commit chart
- D. CSV quote chart

Answer: A

Explanation: Bar charts compare categorical values.

QUESTION 29 - MCQ - 1 MARK

Which function is commonly used to display a Matplotlib chart window?

- A. show()
- B. commit()
- C. execute()
- D. cursor()

Answer: A

Explanation: plt.show() displays the prepared plot.

QUESTION 30 - MCQ - 1 MARK

Which method is used to label the x-axis in PyPlot?

- A. xlabel()
- B. connect()
- C. fetchone()
- D. wrap()

Answer: A

Explanation: xlabel() sets the label for the x-axis.

FILLUP QUESTIONS

QUESTION 31 - FILLUP - 1 MARK

Python is typically an _____ language, while C++ is typically compiled.

Answer: _____

Answer: interpreted | Accepted: interpreted

Explanation: The notes describe Python as interpreted.

QUESTION 32 - FILLUP - 1 MARK

C++ is usually _____ typed, while Python is dynamically typed.

Answer: _____

Answer: statically | Accepted: static|statically

Explanation: C++ requires declared data types in the comparison.

QUESTION 33 - FILLUP - 1 MARK

The process of importing C++ code into Python is called _____.

Answer: _____

Answer: wrapping | Accepted: wrapping|wrapping up

Explanation: Wrapping creates a Python interface for C++ programs.

QUESTION 34 - FILLUP - 1 MARK

SWIG stands for Simplified Wrapper Interface _____.

Answer: _____

Answer: Generator | Accepted: generator

Explanation: SWIG is Simplified Wrapper Interface Generator.

QUESTION 35 - FILLUP - 1 MARK

MinGW allows C++ programs to be compiled and executed dynamically through a _____ program.

Answer: _____

Answer: Python | Accepted: python

Explanation: The notes explain MinGW/g++ execution through Python.

QUESTION 36 - FILLUP - 1 MARK

The Python module used for SQLite database handling is _____.

Answer: _____

Answer: sqlite3 | Accepted: sqlite3

Explanation: sqlite3 is the native module for SQLite.

QUESTION 37 - FILLUP - 1 MARK

The sqlite3 _____ method opens or creates a database file.

Answer: _____

Answer: connect | Accepted: connect|connect()

Explanation: connect() creates the database connection.

QUESTION 38 - FILLUP - 1 MARK

The cursor object is created using the _____ method of a connection.

Answer: _____

Answer: cursor | Accepted: cursor|cursor()

Explanation: connection.cursor() creates a cursor.

QUESTION 39 - FILLUP - 1 MARK

The SQL command is run in Python using cursor._____.()

Answer: _____

Answer: execute | Accepted: execute

Explanation: cursor.execute() executes SQL.

QUESTION 40 - FILLUP - 1 MARK

Changes made to database records are saved permanently using _____().

Answer: _____

Answer: commit | Accepted: commit

Explanation: commit() saves changes.

QUESTION 41 - FILLUP - 1 MARK

The SELECT statement is used to _____ data from a table.

Answer: _____

Answer: retrieve | Accepted: retrieve|fetch|display

Explanation: SELECT is used to retrieve/fetch table data.

QUESTION 42 - FILLUP - 1 MARK

The _____ method fetches all rows from a query result.

Answer: _____

Answer: fetchall | Accepted: fetchall|fetchall()

Explanation: fetchall() returns all rows.

QUESTION 43 - FILLUP - 1 MARK

The fetchone() method returns _____ if there is no row left.

Answer: _____

Answer: None | Accepted: none

Explanation: fetchone() returns None when no row remains.

QUESTION 44 - FILLUP - 1 MARK

Data visualization is the graphical representation of _____ and data.

Answer: _____

Answer: information | Accepted: information

Explanation: Visualization represents information and data graphically.

QUESTION 45 - FILLUP - 1 MARK

Matplotlib is a popular data visualization _____ in Python.

Answer: _____

Answer: library | Accepted: library

Explanation: Matplotlib is a Python visualization library.

QUESTION 46 - FILLUP - 1 MARK

A scatter plot shows data as a collection of _____.

Answer: _____

Answer: points | Accepted: points

Explanation: Scatter plot uses points for two-dimensional values.

QUESTION 47 - FILLUP - 1 MARK

A box plot displays a distribution based on the _____ number summary.

Answer: _____

Answer: five | Accepted: five|5

Explanation: Box plot uses a five-number summary.

QUESTION 48 - FILLUP - 1 MARK

The middle value in a box plot five-number summary is the _____.

Answer: _____

Answer: median | Accepted: median

Explanation: Median is one item in the five-number summary.

QUESTION 49 - FILLUP - 1 MARK

A dashboard translates complex ideas into a simple _____ format.

Answer: _____

Answer: visual | Accepted: visual

Explanation: Dashboard simplifies complex ideas visually.

QUESTION 50 - FILLUP - 1 MARK

The PyPlot function _____() is commonly used to display the graph.

Answer: _____

Answer: show | Accepted: show|plt.show

Explanation: show() displays a Matplotlib graph.

CODING QUESTIONS

QUESTION 51 - CODING - 5 MARKS

Write a function `build_cpp_command(py_file, cpp_name)` that returns the command string to execute a Python wrapper with input mode. Example: `build_cpp_command("pycpp.py", "pali")` should return `"python pycpp.py -i pali"`.

Starter Code:

```
def build_cpp_command(py_file, cpp_name):  
    # write your code here  
    pass
```

Visible Test Cases:

Input: `build_cpp_command("pycpp.py", "pali")`
Expected: `python pycpp.py -i pali`

Input: `build_cpp_command("run.py", "demo")`
Expected: `python run.py -i demo`

Answer: Teacher review / hidden tests

Explanation: Return an f-string or concatenated string in the format: `python <py_file> -i <cpp_name>`

Hidden Test Cases:

```
assert build_cpp_command("main.py", "calc") == "python main.py -i calc"  
assert build_cpp_command("wrapper.py", "pali") == "python wrapper.py -i pali"
```

QUESTION 52 - CODING - 5 MARKS

Write a function `is_palindrome_number(n)` that returns True if the integer reads the same forward and backward, otherwise False.

Starter Code:

```
def is_palindrome_number(n):
    # write your code here
    pass
```

Visible Test Cases:

Input: `is_palindrome_number(232)`
Expected: True

Input: `is_palindrome_number(234)`
Expected: False

Answer: Teacher review / hidden tests

Explanation: Convert the number to string and compare it with its reverse.

Hidden Test Cases:

```
assert is_palindrome_number(121) is True
assert is_palindrome_number(987) is False
assert is_palindrome_number(5) is True
```

QUESTION 53 - CODING - 5 MARKS

Write a function `sqlite_connect_line(db_name)` that returns the Python code line for creating a SQLite connection.

Example: `sqlite_connect_line("Academy.db")` returns `"connection = sqlite3.connect("Academy.db")"`.

Starter Code:

```
def sqlite_connect_line(db_name):
    # write your code here
    pass
```

Visible Test Cases:

Input: `sqlite_connect_line("Academy.db")`
Expected: `connection = sqlite3.connect("Academy.db")`

Answer: Teacher review / hidden tests

Explanation: Return the exact code line as a string.

Hidden Test Cases:

```
assert sqlite_connect_line("school.db") == "connection = sqlite3.connect("school.db")"
```

QUESTION 54 - CODING - 5 MARKS

Write a function `make_insert_student(name, grade, gender, average, birth_date)` that returns an SQL INSERT string using NULL for Rollno.

Starter Code:

```
def make_insert_student(name, grade, gender, average, birth_date):
    # write your code here
    pass
```

Visible Test Cases:

Input: `make_insert_student("Akshay", "B", "M", 87.8, "2001-12-12")`
Expected: `INSERT INTO Student (Rollno,Sname,Grade,gender,Average,birth_date) VALUES (NULL,"Akshay","B","M",87.8,"2001-12-12");`

Answer: Teacher review / hidden tests

Explanation: Use string formatting to build the INSERT statement exactly.

Hidden Test Cases:

```
assert make_insert_student("Aravind", "A", "M", 92.5, "2000-08-17") == "INSERT INTO Student (Rollno,Sname,Grade,gender,Average,birth_date) VALUES (NULL,\"Aravind\", \"A\", \"M\", 92.5, \"2000-08-17\");"
```


QUESTION 55 - CODING - 5 MARKS

Write a function `filter_above_average(records, limit)` that receives a list of tuples (name, average) and returns the names whose average is greater than limit.

Starter Code:

```
def filter_above_average(records, limit):  
    # write your code here  
    pass
```

Visible Test Cases:

Input: `filter_above_average([("Baskar", 75.2), ("Priya", 98.6)], 80)`
Expected: `["Priya"]`

Answer: Teacher review / hidden tests

Explanation: Loop through records and collect names with average > limit.

Hidden Test Cases:

```
assert filter_above_average([("A", 70), ("B", 85), ("C", 90)], 80) == ["B", "C"]
```

QUESTION 56 - CODING - 5 MARKS

Write a function `fetch_one_simulated(rows)` that returns the first row from a list, or None if the list is empty.

Starter Code:

```
def fetch_one_simulated(rows):  
    # write your code here  
    pass
```

Visible Test Cases:

Input: `fetch_one_simulated([(1, "Akshay"), (2, "Aravind")])`
Expected: `(1, "Akshay")`

Input: `fetch_one_simulated([])`
Expected: `None`

Answer: Teacher review / hidden tests

Explanation: Return `rows[0]` if rows is not empty, else None.

Hidden Test Cases:

```
assert fetch_one_simulated([(10, "X")]) == (10, "X")  
assert fetch_one_simulated([]) is None
```

QUESTION 57 - CODING - 5 MARKS

Write a function `count_records(rows)` that returns the number of records returned by a query result list.

Starter Code:

```
def count_records(rows):  
    # write your code here  
    pass
```

Visible Test Cases:

Input: `count_records([(1, "A"), (2, "B"), (3, "C")])`
Expected: `3`

Answer: Teacher review / hidden tests

Explanation: Use `len(rows)`.

Hidden Test Cases:

```
assert count_records([]) == 0  
assert count_records([(1, ), (2, )]) == 2
```

QUESTION 58 - CODING - 5 MARKS

Write a function `choose_chart_type(purpose)` that returns "bar" for comparison, "pie" for parts, "line" for trend, and "scatter" for relationship. Return "unknown" otherwise.

Starter Code:

```
def choose_chart_type(purpose):  
    # write your code here  
    pass
```

Visible Test Cases:

Input: `choose_chart_type("comparison")`
Expected: bar

Input: `choose_chart_type("parts")`
Expected: pie

Answer: Teacher review / hidden tests

Explanation: Use if/elif or a dictionary mapping.

Hidden Test Cases:

```
assert choose_chart_type("trend") == "line"  
assert choose_chart_type("relationship") == "scatter"  
assert choose_chart_type("other") == "unknown"
```

QUESTION 59 - CODING - 5 MARKS

Write a function `five_number_summary(values)` that returns a tuple (minimum, q1, median, q3, maximum). For this assessment, assume the input list has exactly five sorted values.

Starter Code:

```
def five_number_summary(values):  
    # values will contain exactly five sorted numbers  
    # write your code here  
    pass
```

Visible Test Cases:

Input: `five_number_summary([10, 20, 30, 40, 50])`
Expected: (10, 20, 30, 40, 50)

Answer: Teacher review / hidden tests

Explanation: Return `values[0]`, `values[1]`, `values[2]`, `values[3]`, `values[4]`.

Hidden Test Cases:

```
assert five_number_summary([1, 3, 5, 7, 9]) == (1, 3, 5, 7, 9)
```

QUESTION 60 - CODING - 5 MARKS

Write a function `pyplot_line_code(x_values, y_values)` that returns the code string `"plt.plot(x_values, y_values)"`. This checks your understanding of basic PyPlot line plotting syntax.

Starter Code:

```
def pyplot_line_code(x_values, y_values):  
    # write your code here  
    pass
```

Visible Test Cases:

Input: `pyplot_line_code([1, 2, 3], [2, 4, 6])`
Expected: `plt.plot(x_values, y_values)`

Answer: Teacher review / hidden tests

Explanation: Return the exact string requested.

Hidden Test Cases:

```
assert pyplot_line_code([1], [2]) == "plt.plot(x_values, y_values)"
```